

The impact of vegan diet on B-12 status in healthy omnivores: five-year prospective study.

BACKGROUND: There are no long-term prospective studies assessing the impact of the vegan diet on vitamin B-12 (B-12) status. Many vegans take B-12 supplements irregularly or refuse to adopt them at all, considering them to be "unnatural" products. The use of B-12 fortified food may be an alternative. Therefore, we aimed to estimate the long-term effect of a vegan diet on serum B-12 concentrations in healthy omnivore adults, comparing the influence of natural products consumption and B-12 fortified food. **MATERIAL AND METHODS:** A five year prospective study was carried out comprising 20 omnivore healthy adult subjects, who moved to strict vegan diet for 5 years. Ten volunteers followed vegan diet based entirely on natural products, while the remaining ten subjects consumed food fortified in B-12. In all subjects serum vitamin B-12 concentration was determined before and 6, 12, 24 and 60 months after the implementation of the diet. **RESULTS:** A significant decrease ($p < 0.0002$) of serum B-12 concentrations in the whole studied group was noted after 60 months of vegan diet. However, observed changes were in fact limited to the subgroup consuming exclusively natural products ($p < 0.0001$). **CONCLUSIONS:** Transition from omnivore to vegan diet is associated with the risk of vitamin B-12 deficiency. B-12 fortified products might constitute a valuable alternative in vegans refusing to take vitamin supplements.